Compressor: Operation and Maintenance Manual

Applies to: Atlas Copco GA 37, Kaeser CSD 85, Ingersoll Rand R55. Document no. OM-COMPRESSOR-2026.

1. Safety

Lock out and tag out all energy sources before any maintenance. Only trained personnel may work on this machine. Wear the PPE required by the plant safety rules.

2. Sensors and alarm thresholds

An alarm is raised when a value stays above the threshold for at least two consecutive samples. HIGH is a warning; HIGH_HIGH is critical.

Sensor	Location	Unit	Normal	HIGH (warning)	HIGH_HIGH (critical)
temperature	discharge	degC	approx. 85	> 105	> 115
pressure	outlet	bar	approx. 7.5	> 8.5	> 9.5
vibration	motor	mm/s	approx. 2.5	> 5	> 8
current	motor	A	approx. 55	> 75	> 88

3. Alarm response

HIGH (warning): Acknowledge within 15 minutes. Review the trend of the last 2 hours and schedule an inspection within 24 hours. Follow the troubleshooting table for the sensor.

HIGH_HIGH (critical): Reduce load or stop the machine within 5 minutes and notify the shift lead. Open a corrective maintenance order. Do not restart until the root cause is found.

COMM_FAULT: Sensor readings are invalid (quality 'bad'). Check the sensor cable, connector and I/O module. If the fault lasts longer than 4 hours, take manual readings every hour.

4. Troubleshooting

4.1 High temperature (discharge, degC)

Probable causes:

- Clogged oil cooler or aftercooler fins
- Low compressor oil level
- Ambient temperature above 40 degC or poor room ventilation
- Thermostatic bypass valve stuck open

Corrective actions:

1. Clean the cooler fins with compressed air from the inside out.
2. Check the oil level in the sight glass; top up with RotoFluid-46 only.
3. Ensure room ventilation keeps ambient temperature below 40 degC.
4. At HIGH_HIGH the controller trips the compressor automatically. Replace the thermostatic valve kit SP-CMP-TV-46 before restart if the fault repeats.

4.2 High pressure (outlet, bar)

Probable causes:

- Unload pressure set point configured too high
- Minimum-pressure valve stuck
- Downstream consumers or isolation valves closed

Corrective actions:

1. Verify the unload pressure set point is 8.0 bar (load at 7.0 bar).
2. Inspect the minimum-pressure valve; service kit SP-CMP-MPV-02.
3. Never bypass the safety valve; it opens at 10 bar and must be tested every 12 months.

4.3 High vibration (motor, mm/s)

Probable causes:

- Motor-to-air-end misalignment
- Worn coupling element
- Motor bearing damage

Corrective actions:

1. Perform laser shaft alignment (max offset 0.05 mm).
2. Replace the flexible coupling element SP-CMP-CPL-55.
3. Re-grease motor bearings with 20 g of Polyrex EM every 4,000 operating hours.

4.4 High current (motor, A)

Probable causes:

- Supply voltage below tolerance
- Operating pressure above set point
- Air-end wear increasing torque demand

Corrective actions:

1. Measure the supply voltage: 400 V +/- 10% on all three phases.
2. Check the pressure set point (see pressure troubleshooting).
3. If current stays high at normal pressure, schedule an air-end inspection.

5. Maintenance schedule

Interval	Task
Daily	Check outlet pressure and drain condensate
Every 30 days	Clean cooler fins; check oil level
Every 2,000 h	Replace air filter SP-CMP-AF-37
Every 4,000 h	Replace oil, oil filter and separator element
Every 12 months	Test safety valve (opens at 10 bar)